

Monitoring tools should be able to check log file size, parse text for particular string patterns, etc, and perform configured actions. This gives a lot of power to admins to tune their infrastructure monitoring for better control.

Device grouping: This is important for easy management of devices such as firewalls, servers, etc, in specific groups. In some cases, administrators choose to create department wise groups, or a group for each building or floor. They populate these groups with network switches, servers and desktops pertaining to that department or floor. In a growing infrastructure, this feature is very important.

Alert management: Merely monitoring a network is not enough. A good tool should let admins produce alerts. For example, if the CPU of a critical server crosses 90 per cent of usage, or if a firewall is dropping multiple packets in a row, it should create a trouble ticket, and email, or optionally send a short message to the admin's mobile phone. Almost all tools provide such facilities today to enhance their usefulness; however, admins should look into the configurability and facilities available in alert management, prior to selecting the proper tool.

Customisable Web dashboard: A good monitoring tool should let admins access its statistics over a Web interface. Besides, the Web interface must be customisable to let them decide what should be on the dashboard's front page. Modern tools provide widgets that are small screen sections or windows, which can show monitoring statistics of the admin's choice and can be moved or removed.

Integrating with helpdesk: Recording of events that are the result of threshold violation is very important and should be an automated process. The monitoring tool should provide the necessary hooks or connectors so that the trouble ticket/helpdesk system can be easily connected. Monitoring of events and the applicable actions should result in a trouble ticket. This helps decide how much manpower ought to be utilised to address those events and intelligent action can be taken based on that data.

Report generation: All monitoring tools today provide some level of report generation, which is based on the date, time, etc. However, a detailed report that is device-specific or event-specific is really essential for an admin. For example, a report generator should be able to drill down into a particular event such as a TCP timeout on a particular server, and provide historic occurrences of that event for that server. These levels of granular details help administrators establish a co-relation between the event and its root cause.

The following is a list of new features found in commercial monitoring tools; however, the open source world will surely catch up in the days to come.

Plug-in API support: While a few open source tools do provide this, there is still a scope for improvement. API calls of the monitoring engine can be exposed in a secure way, so that developers can write their own plug-ins. This is especially important when there is a new network device or

Table 1

Layer	Feature examples
Application	App-specific log check App run/hung state App performance
Database	Connectivity Performance Query optimisation and response
Operating system	Task performance Security events OS service run/hung state
Web	URL query string monitoring Web services response Web service up/down
Hardware	CPU utilisation Memory utilisation Disk quota utilisation
Network	Packet drops and anomalies Network utilisation Bandwidth usage

software application in the market that must be monitored.

Trend analysis: The network or server monitoring industry is rapidly moving away from the preventive to the pro-active mode. Administrators want to know historic trends of problems and make a judgment in terms of the corrective actions to be taken today, to prevent problems that might happen tomorrow. For example, continuously high CPU utilisation on a MySQL server over a period suggests that one or more stored procedures are either not optimised or misbehaving. This can be related to the application that uses those procedures. Thus, if that application is expected to be used more, an analysis of the trends can tell admins that the MySQL server is going to run into trouble.

Security monitoring: Very soon, no monitoring tool will be useful unless it supports cyber security monitoring. Attacks happening at Layer 2 and 3, as well as application-based security problems at Layer 7, should be trapped and reported by a good monitoring tool. This functionality is available in a few commercial tools; however, incorporating Snort along with Nagios or any other monitoring tool can prove to be a powerful security monitoring solution.

Nagios, Zenoss and Zabbix

So let's talk about the three famous open source monitoring tools—Nagios, Zenoss and Zabbix, and compare them. While there are many features to compare, we will discuss only those that matter the most to mid-scale IT infrastructure management.

Nagios: This is a famous first generation network monitoring tool and is used in all Linux distros. Developed in C and PHP, it supports multiple flavours of open source backend databases, as well as the legacy flat file structure.

Zenoss: Written using Python scripting, Zenoss provides